PharmaZen

Software Requirements Specification

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This document is intended for the use of the PharmaZen project team and its authorized stakeholders.

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SECTION 1 · INTRODUCTION

1 Introduction

This document establishes the complete Software Requirements Specification (SRS) for the **PharmaZen** website — an online pharmacy and health-product platform. It defines all functional and non-functional requirements to guide the development, testing, and delivery lifecycle.

1.1 Purpose

The purpose of this SRS is to provide a detailed and agreed-upon description of the PharmaZen website system. It serves as the foundational reference for all project stakeholders — including developers, designers, QA engineers, and business owners — ensuring a shared understanding of what the system must accomplish.

1.2 Intended Audience

This document is intended for the following parties:

- **Development Team** — Front-end and back-end engineers responsible for building the system
- **UI/UX Designers** — Responsible for designing user interfaces that satisfy the stated requirements
- **QA Engineers** — Responsible for validating implemented features against this specification
- **Project Managers** — Overseeing schedule, scope, and delivery milestones
- **Business Stakeholders** — PharmaZen owners and executives reviewing product direction
- **Regulatory Reviewers** — Ensuring compliance with pharmaceutical and data regulations

1.3 Intended Use

This SRS shall be used as:

- A contractual reference between the development team and business stakeholders
- A baseline for test plan development and acceptance criteria
- A guide for identifying and managing change requests throughout the project lifecycle
- A reference document for onboarding new team members

1.4 Product Scope

PharmaZen is a comprehensive e-commerce website specializing in pharmaceutical products, health supplements, personal care items, and related medical goods. The platform will enable customers to browse products, upload prescriptions, place orders, and manage their health profiles.

Core capabilities of the platform include:

- Online pharmacy storefront with product catalog and search
- Prescription upload, verification, and management
- Secure user registration, authentication, and profile management
- Order placement, real-time tracking, and order history
- Admin dashboard for inventory, order, and user management
- Integration with third-party payment gateways
- Pharmacist consultation request and messaging system

1.5 Risk Definitions

The following risk severity levels are used throughout this document:

Level	Category	Description
Critical	Business / Legal	Non-compliance with pharmaceutical regulations or data protection laws that could result in shutdown or legal action
High	Security	Breaches or vulnerabilities that expose patient or payment data
Medium	Operational	System downtime or feature failures that degrade user experience significantly
Low	Performance	Minor slowdowns or UI inconsistencies that do not block core functionality

SECTION 2 · OVERALL DESCRIPTION

2 Overall Description

This section provides a high-level overview of the PharmaZen system, including its user base, operating environment, constraints, and assumptions that govern its design and development.

2.1 User Classes and Characteristics

The PharmaZen platform serves four primary user classes, each with distinct access levels and interaction needs:

Guest Visitor	Unauthenticated users who can browse the product catalog, view product details and prices, and create an account. Cannot place orders or upload prescriptions.
Registered Customer	Authenticated users who can manage profiles, upload prescriptions, place and track orders, access order history, and contact pharmacists. Primary revenue-generating user class.
Pharmacist / Staff	Licensed healthcare professionals who verify prescriptions, respond to consultation requests, manage product listings, and oversee medication safety flags.
System Administrator	Technical and business administrators who manage all platform operations including user accounts, inventory, promotions, reports, and system configurations.

2.2 User Needs

Based on user class analysis, the following core needs have been identified:

2.2.1 Registered Customers

- Easily search, filter, and find pharmaceutical and health products
- Securely upload and manage prescriptions for prescribed medication orders
- Complete purchases quickly with multiple payment options
- Track orders in real time from placement to delivery
- Access a trusted pharmacist for health-related queries

2.2.2 Pharmacists / Staff

- Efficiently review and approve or reject uploaded prescriptions
- Flag medications that require counseling or special handling
- Respond to customer consultation requests in a timely manner

- Manage and update product listings with accurate medical information

2.2.3 System Administrators

- Monitor platform health, performance, and transaction logs
- Manage product inventory levels, categories, and pricing
- Handle user disputes, returns, and policy enforcement
- Generate business intelligence reports and analytics

2.3 Operating Environment

The PharmaZen website will operate in the following technical environment:

Platform Type	Web application accessible via modern desktop and mobile browsers
Frontend	React.js (Vite) — responsive single-page application targeting screens from 320px to 2560px wide
Backend	Node.js with Express.js REST API, or equivalent server-side framework
Database	PostgreSQL hosted on Neon Tech (serverless); Redis for session/cache management
Cloud Hosting	Vercel (frontend and serverless functions); automatic CI/CD from Git repository
CDN	Vercel Edge Network for static assets, images, and global performance
Browser Support	Chrome 100+, Firefox 100+, Safari 15+, Edge 100+ on desktop and mobile
Payment Gateway	Create mock bkash server for token pass testing, actual Sandbox requires trade license
SSL/TLS	HTTPS enforced across all pages and API endpoints

2.4 Constraints

- **Regulatory Compliance:** The platform must comply with applicable pharmaceutical regulations and digital health guidelines in the operating jurisdiction.
- **Data Privacy:** All user health and personal data must be handled in accordance with applicable data protection regulations (e.g., GDPR, HIPAA or local equivalents).
- **Prescription Verification:** Medications classified as prescription-only must not be dispensed without a valid, verified prescription from a licensed pharmacist.
- **Payment Security:** All payment processing must meet PCI DSS Level 1 compliance standards.
- **Technology Stack:** The development team is constrained to use the agreed-upon technology stack to ensure maintainability and team expertise alignment.

- **Budget & Timeline:** Development must be delivered within the agreed project budget and timeline as defined in the project charter.
- **Third-Party APIs:** Features relying on external APIs (e.g., delivery tracking, payment) are subject to the availability and terms of those third-party providers.

2.5 Assumptions

- All users accessing the platform are assumed to have access to a stable internet connection.
- The platform will initially target a single jurisdiction/country; multi-country expansion is out of scope for v1.0.
- Licensed pharmacists reviewing prescriptions will be pre-vetted and onboarded by PharmaZen management.
- Payment gateway providers will maintain their service agreements and uptime SLAs.
- Product catalog and inventory data will be initially seeded by the PharmaZen operations team.
- Users are responsible for providing accurate health and prescription information; the platform is not liable for fraudulent submissions.
- Third-party delivery partners will be responsible for last-mile logistics, and PharmaZen will integrate only with their tracking APIs.

SECTION 3 · SYSTEM REQUIREMENTS

3 Requirements

3.1 Functional Requirements

Functional requirements define the specific behaviors and capabilities the PharmaZen system must provide.

3.1.1 User Registration & Authentication

ID	Requirement	Description	Priority
FR-01	**User Registration**	Users can register using an email address and password, or via social login (Google/Facebook OAuth 2.0). Email verification is required before account activation.	High
FR-02	**User Login**	Registered users can log in with valid credentials. Failed login attempts (5+) lock the account for 15 minutes with an email notification.	High
FR-03	**Password Reset**	Users can request a password reset link via email. Reset links expire after 30 minutes.	High
FR-04	**Multi-Factor Auth**	Users can optionally enable MFA (TOTP or SMS) for enhanced account security.	Medium
FR-05	**Profile Management**	Authenticated users can update personal details (name, address, phone), medical profile, and communication preferences.	High
FR-06	**Account Deactivation**	Users can request account deactivation; data is retained for 90 days before deletion per data retention policy.	Medium

3.1.2 Product Catalog & Search

ID	Requirement	Description	Priority
FR-07	**Product Listings**	All products must display name, image, category, price, dosage (if applicable), availability status, and a manufacturer description.	High
FR-08	**Search Functionality**	Users can search products by name, category, brand, or active ingredient. Search results appear within 2 seconds.	High
FR-09	**Advanced Filters**	Users can filter results by price range, category, brand, prescription requirement, and in-stock status.	Medium

ID	Requirement	Description	Priority
FR-10	**Product Detail Page**	Each product has a dedicated page with full description, images, dosage information, side effects, and related products.	High
FR-11	**Wishlist**	Authenticated users can save products to a personal wishlist and receive restock alerts.	Low
FR-12	**Product Reviews**	Verified purchasers can submit ratings (1–5 stars) and written reviews. Reviews are moderated before publishing.	Medium

3.1.3 Prescription Management

ID	Requirement	Description	Priority
FR-13	**Prescription Upload**	Users can upload prescriptions in PDF, JPEG, or PNG format (max 10 MB) during or after checkout for prescription-required items.	High
FR-14	**Prescription Review**	Pharmacist staff can view, approve, or reject uploaded prescriptions with mandatory comment fields for rejections.	High
FR-15	**Prescription History**	Users can access a history of all submitted prescriptions and their current statuses.	Medium
FR-16	**Prescription Expiry**	The system tracks prescription validity dates and alerts users when prescriptions are about to expire (7-day warning).	Medium
FR-17	**Auto-Fill from Rx**	Upon prescription approval, the associated products are automatically flagged as fulfillable in the user's cart.	Low

3.1.4 Shopping Cart & Checkout

ID	Requirement	Description	Priority
FR-18	**Shopping Cart**	Users can add, remove, and update quantities of products in a persistent cart (saved across sessions for authenticated users).	High
FR-19	**Checkout Process**	Checkout flow includes delivery address selection, shipping method, payment, and order summary review before confirmation.	High
FR-20	**Payment Gateway**	The system supports payment via credit/debit card (Stripe) and digital wallets. All transactions are processed over HTTPS.	High

ID	Requirement	Description	Priority
FR-21	**Promo Codes**	Users can apply promotional codes at checkout. The system validates and applies the appropriate discount automatically.	Medium
FR-22	**Order Confirmation**	Upon successful payment, the user receives an on-screen confirmation and an email receipt with order details.	High
FR-23	**Guest Checkout**	Guest users may complete a purchase without creating an account. Post-purchase, they are prompted to register.	Low

3.1.5 Order Management

ID	Requirement	Description	Priority
FR-24	**Order History**	Authenticated users can view all past orders with status, products, total cost, and delivery details.	High
FR-25	**Order Tracking**	Users receive real-time tracking updates via in-app notifications and email when order status changes.	High
FR-26	**Order Cancellation**	Users can cancel orders within 1 hour of placement if the order has not been dispatched.	Medium
FR-27	**Returns & Refunds**	Users can submit a return request within 7 days for eligible non-pharmaceutical items. Refunds are processed within 5–7 business days.	Medium
FR-28	**Reorder**	Users can reorder previous orders with a single click, adding all eligible items back to cart.	Low

3.1.6 Pharmacist Consultation

ID	Requirement	Description	Priority
FR-29	**Consultation Request**	Users can submit a consultation request with a description of their query and optionally attach relevant documents.	Medium
FR-30	**Pharmacist Messaging**	An in-platform secure messaging thread is created between the user and assigned pharmacist for each consultation.	Medium
FR-31	**Consultation History**	Users can access all past consultations with timestamps and pharmacist responses.	Low

3.1.7 Admin Dashboard

ID	Requirement	Description	Priority
FR-32	**Inventory Management**	Admins can add, edit, deactivate products, and set low-stock alert thresholds.	**High**
FR-33	**User Management**	Admins can view, suspend, or delete user accounts, and manage pharmacist onboarding.	**High**
FR-34	**Order Management**	Admins can view all orders, update statuses, initiate refunds, and flag disputes.	**High**
FR-35	**Analytics & Reporting**	Admins have access to sales reports, traffic analytics, prescription volumes, and user growth dashboards.	**Medium**
FR-36	**Content Management**	Admins can manage homepage banners, promotional sections, and featured product lists.	**Medium**
FR-37	**Audit Log**	All admin actions are recorded in an immutable audit log with timestamp and admin ID.	**High**

3.2 Non-Functional Requirements

Non-functional requirements define the quality attributes and operational constraints the PharmaZen system must satisfy.

3.2.1 Performance

ID	Requirement	Description	Priority
NFR-01	**Page Load Time**	All public-facing pages must load within 3 seconds under normal load conditions (up to 500 concurrent users).	High
NFR-02	**Search Response**	Product search queries must return results within 2 seconds for 95% of requests.	High
NFR-03	**Checkout Throughput**	The checkout process must support at least 100 concurrent transactions without degradation in response time.	High
NFR-04	**Scalability**	The system must horizontally scale to support up to 10,000 concurrent users during peak periods (e.g., promotional events).	Medium

3.2.2 Security

ID	Requirement	Description	Priority
NFR-05	**Data Encryption**	All sensitive data (passwords, health records, payment info) must be encrypted at rest (AES-256) and in transit (TLS 1.2+).	High
NFR-06	**Auth Security**	Passwords must be hashed using bcrypt (minimum cost factor 12). Plain-text passwords must never be stored.	High
NFR-07	**Session Management**	User sessions must expire after 30 minutes of inactivity. JWT tokens must have a maximum lifespan of 24 hours.	High
NFR-08	**Input Validation**	All user inputs must be validated and sanitized server-side to prevent SQL injection, XSS, and CSRF attacks.	High
NFR-09	**PCI DSS Compliance**	Payment processing must comply with PCI DSS Level 1 standards. Card data must never be stored on PharmaZen servers.	High
NFR-10	**Vulnerability Scan**	The platform must undergo automated vulnerability scans (OWASP Top 10) before each major release.	Medium

3.2.3 Reliability & Availability

ID	Requirement	Description	Priority
NFR-11	**Uptime SLA**	The platform must maintain a minimum uptime of 99.5% measured on a monthly basis (excluding scheduled maintenance windows).	High
NFR-12	**Maintenance Windows**	Maintenance windows must be communicated to users at least 24 hours in advance and conducted during off-peak hours.	Medium
NFR-13	**Data Backup**	Full database backups must run daily; incremental backups every 6 hours. Backups must be retained for 30 days.	High
NFR-14	**Disaster Recovery**	Recovery Point Objective (RPO) of 6 hours and Recovery Time Objective (RTO) of 4 hours must be maintained.	High
NFR-15	**Failover**	The system must implement automatic failover for database and application servers to minimize downtime.	Medium

3.2.4 Usability & Accessibility

ID	Requirement	Description	Priority
NFR-16	**Responsive Design**	All pages must be fully functional and visually consistent across screen sizes from 320px (mobile) to 2560px (desktop).	High
NFR-17	**WCAG Compliance**	The platform must conform to WCAG 2.1 Level AA accessibility standards, including screen reader support and keyboard navigation.	High
NFR-18	**Error Messages**	All error messages must be user-friendly, specific, and provide actionable guidance. Technical error codes must not be exposed to end users.	Medium
NFR-19	**Onboarding**	New users should be able to complete registration and place their first order within 5 minutes without external assistance.	Medium

3.2.5 Maintainability & Compliance

ID	Requirement	Description	Priority
NFR-20	**Code Standards**	All code must follow the agreed-upon coding standards and pass automated linting checks before merging to the main branch.	Medium
NFR-21	**Documentation**	All APIs must be documented using OpenAPI 3.0 specification. In-code documentation must be maintained for all modules.	Medium

ID	Requirement	Description	Priority
NFR-22	**Data Retention**	User health and order data must be retained for a minimum of 5 years in compliance with applicable healthcare regulations.	**High**
NFR-23	**GDPR / Privacy**	Users must be able to request export or deletion of their personal data. A privacy policy and cookie consent banner must be displayed.	**High**
NFR-24	**Audit Logging**	All access to sensitive data (prescriptions, health records) must be logged with user ID, timestamp, and action type.	**High**

Document Control

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This document is subject to change management procedures. Any modifications must be reviewed, approved, and version-controlled. Changes will be tracked in the project change log.